OFFICIAL BRAND CATALOGUE • POWER

EMI SOLUTIONS (EMIS)

EMI Solutions Pvt. Ltd. (EMIS Global)

EMI / EMC Filters, Feedthrough Components, Power Quality & EMP Surge Protection



OFFICIAL CHANNEL PARTNER & SUPPLIER

ENGINEERING OVERVIEW & TECHNOLOGY HIGHLIGHTS

EMI Solutions Pvt. Ltd. (EMIS) is a global leader and premier Indian manufacturer specializing in Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC) compliance solutions. With cutting-edge testing facilities and comprehensive engineering, EMIS provides total electromagnetic noise suppression across 50 Hz to 40 GHz: from single, two, and three-phase power line noise filters, IEC power inlet modules, and feedthrough capacitors, to advanced dV/dt and sine wave power quality output filters, military-grade TEMPEST / HEMP chamber filters, and high-capacity surge protection devices.

- Broad Spectrum EMI/EMC Suppression: Solves conducted and radiated EMI challenges from 50 Hz up to 40 GHz across commercial, medical, and defense domains
- Power Quality & Harmonic Mitigation: Tuned dV/dt motor protection filters, sinusoidal output filters, and passive harmonic filters (THDi < 5% / 8%, IEEE 519 compliant)
- Advanced Surge Protection & Spark Quenching: Type 1+2 / Type 2 AC & DC surge protectors (SSP/TSP/DSP series), RC spark quenchers, and high-speed Gigabit RJ45 network protectors
- Comprehensive Power Line Filters: Single-phase (MF300/MF310 IEC inlets, RN/RD series), two-phase (MF422), and heavy-duty three-phase delta/wye filters up to 1200A
- Military & Defense-Grade Protection: Certified TEMPEST SDIP-27 filters, MIL-STD-461E/F power line filters, MIL-STD-188-125 HEMP control/data line filters, and hospital MRI RF shielding

EMI / EMC POWER LINE FILTERS



Single Phase IEC Inlet Power Filters (MF300 / MF310 / MF320 Series)

IEC Inlet

Chassis-mount IEC 320 C14 / C20 power entry filters with integrated EMI noise suppression, optional fuse holder drawer, DPST rocker power switch, and medical low-leakage (<5µA) variants.

• Voltage: 250VAC, 50/60Hz • Current: 1A to 20A • IEC 320 C14 & C20 standard • Snap-in & Screw mount • Medical Grade M low leakage



Single Phase Power Line Noise Filters (RN / RD Series)

Single Phase

Compact chassis and PCB-mount RFI noise suppression filters designed for SMPS power supplies, instrumentation, industrial automation controllers, and sensitive electronics.

• Operating Voltage: 250VAC / 300VAC • Current Rating: 1A to 100A • High differential & common mode attenuation • Fast-on / screw / wire lead terminals



Common Mode Chokes & Inductors (CMC71 / CMC73 / CMC74 Series)

CMC Chokes

High-permeability toroidal core common-mode chokes offering exceptional noise suppression across wide frequency ranges up to 400Hz for PCB and power supplies.

• Voltage: 250V to 600VAC / 850VDC • Current: 0.25A to 64A • Operating Frequency: DC to 400Hz • Through-hole PCB & vertical mounting



Two Phase Power Line EMI Filters (MF422 Series)

Two Phase

Specialized dual-phase (phase-to-phase) EMI filters engineered for 2-phase industrial machinery, dual-phase welding equipment, and commercial appliances.

• Operating Voltage: 440VAC / 480VAC 50/60Hz • Current: 5A to 100A • Dual-stage high noise attenuation • Finger-safe touch-proof terminal blocks



Three Phase Single Stage & 4-Wire Line Filters (MF423-3W & MF-423-4W Series)

Three Phase 3W/4W

Industrial three-phase Delta (3-wire) and Star/Wye with Neutral (4-wire) RFI noise filters designed for motor drives, CNC machines, and factory power distribution.

• Operating Voltage: 440VAC / 520VAC • Current: 5A to 1200A • Finger-safe IP20 touch terminals • Mitigates common & differential mode feedback



Three Phase Dual Stage & High Voltage Filters (MF-423-3-2D & MF423-HV Series)

Dual Stage & HV

Multi-stage high-attenuation line filters and 690VAC heavy-duty filters designed for large Variable Frequency Drives (VFDs), solar central inverters, and wind turbines.

• Operating Voltage: 440VAC / 520VAC / 690VAC • Current: up to 1200A • Extreme attenuation: 150 kHz to 30 MHz • Book-style & compact chassis packaging



DC Power Line & EV Fast Charging EMI Filters (MF-610 & 450-EVX Series)

DC & EV Fast Charge

High-performance DC noise suppression filters engineered for solar photovoltaic inverters, telecom DC power plants, battery energy storage systems (BESS), and electric vehicle fast chargers.

• Operating Voltage: Up to 1200VDC • Current Rating: 10A to 800A • Low power loss & high thermal stability • Chassis-mount with copper busbar / terminal block

FEEDTHROUGH COMPONENTS



Power Feedthrough Capacitors (FTC Series)

Feedthrough FTC

Threaded cylindrical brass and nickel-plated feedthrough capacitors providing ultra-low self-inductance and continuous RF noise filtering up to multi-gigahertz frequencies.

• Capacitance: 10 pF to 4.7 µF • Operating Voltage: Up to 1000VDC / 440VAC • Resonance Frequency: >1 GHz • Hermetically sealed brass/nickel-plated body



Bulkhead Feedthrough Filters (FTF Series)

Feedthrough FTF

Integrated Pi, T, and C low-pass feedthrough filter assemblies engineered for shielded room bulkheads, RF microwave cabinets, radar installations, and defense power lines.

• Current Rating: 5A to 300A • Operating Voltage: 250VAC / 500VDC • High insertion loss: > 70 dB @ 1 GHz • Bolt-in & solder mounting styles

POWER QUALITY & HARMONIC SOLUTIONS



dV/dt Output Motor Protection Filters (TPQDR-1 Series)

dV/dt Output Filter

Motor protection filters installed on VFD inverter output to limit voltage rate-of-rise (dV/dt < 500V/µs), suppressing peak motor terminal overvoltages and cable reflection spikes.

• Motor Voltage: 380V - 500VAC • Motor Power: 1.5 kW to 630 kW • Limits dV/dt < 500V/µs • Prevents motor winding insulation breakdown • Enables motor cable lengths up to 150m

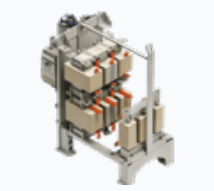


Sine Wave Inverter Output Filters (SINEFILTER_HD Series)

Sine Wave Filter

Heavy-duty LC low-pass sine wave filters transforming PWM drive output into a clean sinusoidal voltage waveform, completely eliminating acoustic motor hum and bearing currents.

• Operating Voltage: 400V / 500VAC 50/60Hz • Switching Frequency: 2 kHz to 8 kHz • Residual ripple voltage: < 5% • Supports ultra-long motor cables > 500m • Eliminates high-frequency bearing fluting



Passive Harmonic Filters (PHF & PHF2 Series)

THDi < 5% / 8%

Tuned passive LC harmonic mitigation filters designed to eliminate 5th, 7th, 11th, and 13th harmonics from 3-phase rectifier inputs, achieving compliance with IEEE 519 standards.

• Total Harmonic Current Distortion (THDi): < 5% or < 8% • Voltage: 380V - 480VAC 50/60Hz • Power Factor: > 0.98 at rated load • IEEE 519-2022 compliant • Current: 10A to 500A

MILITARY GRADE FILTERS & EMP PROTECTION

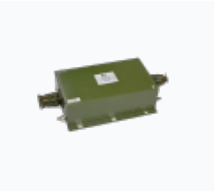


TEMPEST Facility Power Line Shielding Filters (MF710 Series)

TEMPEST SDIP-27

Severe-duty shielded room and data center power line filters meeting NATO TEMPEST SDIP-27 and MIL-STD-188-125, attenuating cyber eavesdropping and RF electromagnetic emissions.

• Attenuation: >100 dB from 14 kHz to 10 GHz • Current: 16A to 1200A Single & 3-Phase • TEMPEST SDIP-27 Level A compliant • Welded RF-tight enclosure with RF mesh gasket



MIL-STD-461E/F Defense Power Line Filters (MF423-3-3D-M Series)

MIL-STD-461

High-reliability defense-grade three-phase power filters tested and certified for CE101 and CE102 conducted emissions compliance on naval, airborne, and military vehicle electronics.

• Standards: MIL-STD-461E / MIL-STD-461F • Voltage: 440VAC 50/60/400Hz • Current: 10A to 400A • Heavy shock & vibration resistant housing



HEMP Control & Signal Line Filters (HEMP-CLF & HEMP-TLF Series)

MIL-STD-188-125

Single-pair and multi-pair control and analog/digital telephone line EMP filters providing high-energy transient suppression and continuous RF attenuation against nuclear HEMP threats.

• MIL-STD-188-125-1 & 2 compliant • Voltage: 24VDC to 250VAC • High surge current withstand (E1 & E2 pulses) • Feedthrough bulkhead mounting



High-Speed Data Line & Ethernet EMP Filters (RJ45 & RS-422 Series)

Ethernet & Data EMP

High-speed Cat5e/Cat6 Ethernet and RS-422/RS-485 serial data line surge and EMP filters protecting defense command networks, radar feeds, and secure communications.

• Data Rate: Gigabit 10/100/1000 Mbps & RS-422/485 • Surge Handling: IEC 61000-4-5 & MIL-STD-188-125 • Low insertion loss on high-speed data • Cast aluminum shielded enclosure



Medical MRI Room RF Shielding Filters (MF420-2-2CF-M Series)

MRI RF Shielding

Dedicated RF power line suppression filter engineered for hospital MRI scanning rooms and sensitive diagnostic imaging facilities, eliminating RF noise artifacts in MRI imagery.

• Attenuation: >100 dB from 10 MHz to 1 GHz • Operating Voltage: 250VAC / 440VAC • Low earth leakage current • Non-magnetic and RF-sealed enclosure

SURGE PROTECTORS & SPARK QUENCHERS



Single Phase DIN-Rail Surge Protectors (SSP2206 / SSP2406 Series)

Single Phase SPD

Pluggable Type 1+2 and Type 2 modular DIN-rail surge protective devices engineered for 230VAC single-phase power distribution panels and control cabinets.

• Voltage: 230VAC / 275VAC • Discharge Current (In): 20kA | (Imax): 40kA • Visual green/red status indicator • Remote signalling alarm dry contact • DIN-rail 35mm mount



Three Phase Industrial Surge Protectors (TSP3206 / TSP3406 / TSP4206 Series)

Three Phase SPD

High-capacity Type 1+2 / Type 2 three-phase surge arresters designed for main low-voltage switchboards, industrial motor control centers, and commercial facilities.

• Voltage: 440VAC (3P+N / 3P) • Maximum Discharge Current (Imax): up to 65kA (8/20 µs) • Fast nanosecond response time (<25ns) • Thermal disconnection protection • Pluggable replaceable varistor cartridges



Three Phase RC Spark Quenchers & Snubbers (T-RCF & TSP4083 Series)

Spark Quencher

Precision resistor-capacitor (RC) snubber networks designed to extinguish electrical arc transients and contact bounce caused by switching relays, contactors, and solenoids.

• Operating Voltage: Up to 440VAC 50/60Hz • Suppresses high inductive back-EMF spikes • Extends contactor contact life • Chassis and DIN rail mounting options



DC Solar Photovoltaic & Battery Surge Protectors (DSP406 Series)

DC SPD 1000V

Photovoltaic and DC power bus surge protective device engineered to clamp lightning and switching overvoltages on solar combiner boxes, inverters, and battery storage systems.

• Rated Voltage (Ucpv): 600VDC / 1000VDC • Discharge Current (Imax): 40kA (8/20 µs) • High-energy metal oxide varistors (MOV) • DIN rail plug-in module



Network & Gigabit Ethernet Signal Surge Protectors (NSPRJ45 Series)

RJ45 Network SPD

High-speed transient voltage surge suppressor equipped with RJ45 shielded connectors, designed to protect IP CCTV cameras, VoIP phones, switches, and PoE equipment from lightning induced surges.

• Connector: RJ45 Shielded (In/Out) • Data Rate: 10/100/1000 Mbps Gigabit • PoE / PoE+ compatible • Fast response time: < 1 ns • Rugged metal ground lug for earth bonding

OFFICIAL COMMERCIAL QUOTATION & TECHNICAL SUPPORT

To request an instant pricing quotation, OEM warranty validation, or technical submittals for this brand:

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